

autonomy_light interface

autonomy_light 는 하나의 ROS 2 node다. Super-LIO의 global-frame point cloud와 LiDAR odometry를 소비하고, robot-centric elevation map을 발행한다.

Direction	Topic	Type	Frame
input	/lio/odom	nav_msgs/msg/Odometry	odom or map → LiDAR
input	/lio/cloud_world	sensor_msgs/msg/PointCloud2	odom or map
output	/autonomy_light/odom	nav_msgs/msg/Odometry	global → base_link
output	/autonomy_light/path	nav_msgs/msg/Path	global
output	/autonomy_light/height_map_data	autonomy_light/msg/HeightMap	base_link_gravity
output	/autonomy_light/height_map_quality	autonomy_light/msg/HeightMapQuality	base_link_gravity
output	/autonomy_light/height_map	sensor_msgs/msg/PointCloud2	base_link_gravity
output	/autonomy_light/live_map	sensor_msgs/msg/PointCloud2	global

TF is odom|map → base_link → base_link_gravity, plus static base_link → lidar_link. A saved map selects Super-LIO relocation_node and therefore uses map; normal operation uses odom. No world frame and no locally generated map → odom correction exist.

HeightMap.data is reference_height - terrain_height, clamped by height_map_distance. A quality mask marks unobserved cells explicitly: valid=0 means data contains only the configured unknown placeholder.

Cyclone DDS transport

Set height_map_output.transport to cyclone_dds for a direct DDS output, or both to retain the ROS 2 outputs during migration. The installed IDL is resources/idl/HeightMap.idl and defines exactly:

```
module core_dds { struct HeightMap { sequence<float> data; }; };
```

The configurable default is domain 1, topic height_map, type core_dds::HeightMap, best-effort keep-last history depth 1. Its sole field has the same row-major cell order, length and unknown convention as ROS HeightMap.data; it intentionally carries no header, geometry or quality channels. Consumers must share the grid configuration and read the ROS quality stream when that metadata is required.